

10.1-10.2

$$\left[-\frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) + \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \right] R_{E,l}(r) = E R_{E,l}(r) \quad (10.2)$$

central potentials go from $r = 0 \rightarrow r \rightarrow \infty$, what happens at $r = 0$? For the SHO, it was $-\infty, \infty$ so we could let it go to zero and be done.

We make an approximation - we can't have a delta function at zero.

$$V_{\text{eff}}(r) = \frac{l(l+1)\hbar^2}{2\mu r^2} + V(r) \quad (10.12)$$

The $l(l+1)/2\mu r^2$ term, known as the **centrifugal barrier**, increases with increasing l and tends to keep the particle away from the origin, producing the small- r behavior of the wave functions (see Fig. 10.1).

Wave functions vanish at the origin like the spherical harmonics did at the poles.

10.2

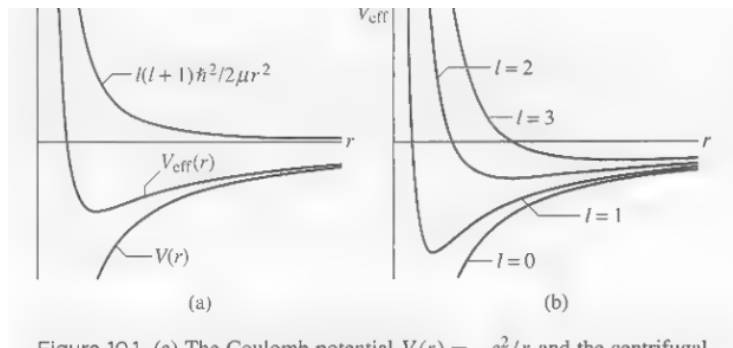


Figure 10.1. (a) The Coulomb potential $V(r) = -e^2/r$ and the centrifugal

$$E = -\frac{\mu Z^2 e^4}{2\hbar^2 (1 + l + n_r)^2} \quad (10.30)$$

Since l and n_r are both integers that are greater than or equal to zero, we define the principal quantum number n by

$$l + 1 + n_r = n \quad (10.31)$$

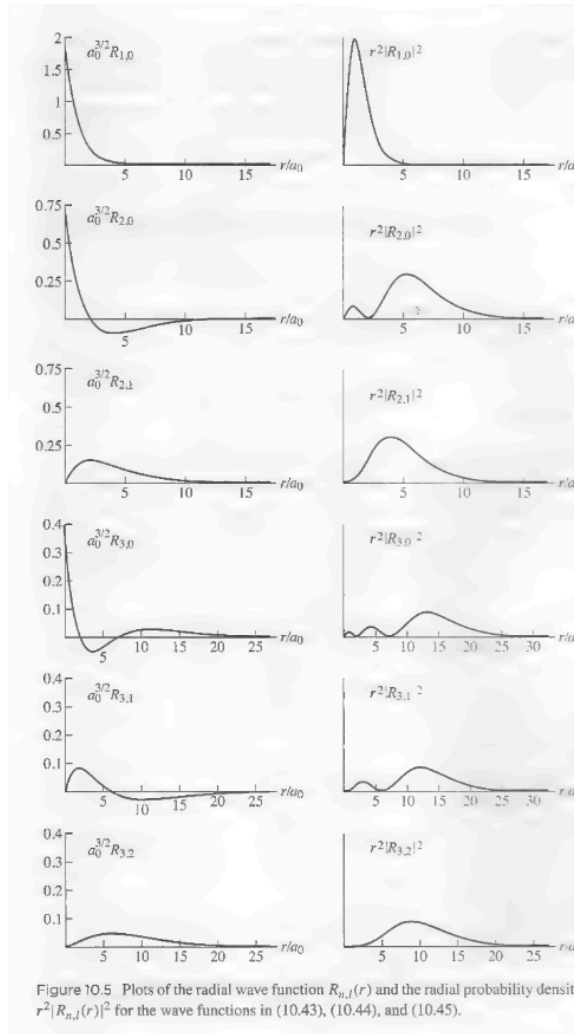
Thus in terms of the principal quantum number

$$E_n = -\frac{\mu Z^2 e^4}{2\hbar^2 n^2} \quad n = 1, 2, 3, \dots \quad (10.32)$$

$$\frac{e^2}{\hbar c} = \alpha \quad (10.33)$$

a dimensionless quantity whose value is approximately 1/137 and is known as the **fine-structure constant**, for reasons that will become apparent in Chapter 11.⁶ In terms of α , the allowed energies are given by

$$E_n = -\frac{\mu c^2 Z^2 \alpha^2}{2n^2} \quad (10.34)$$



What is the Ryberg

There are l degenerate states